Lecture 10: Classic embeddings

PSYC 51.17: Models of language and communication

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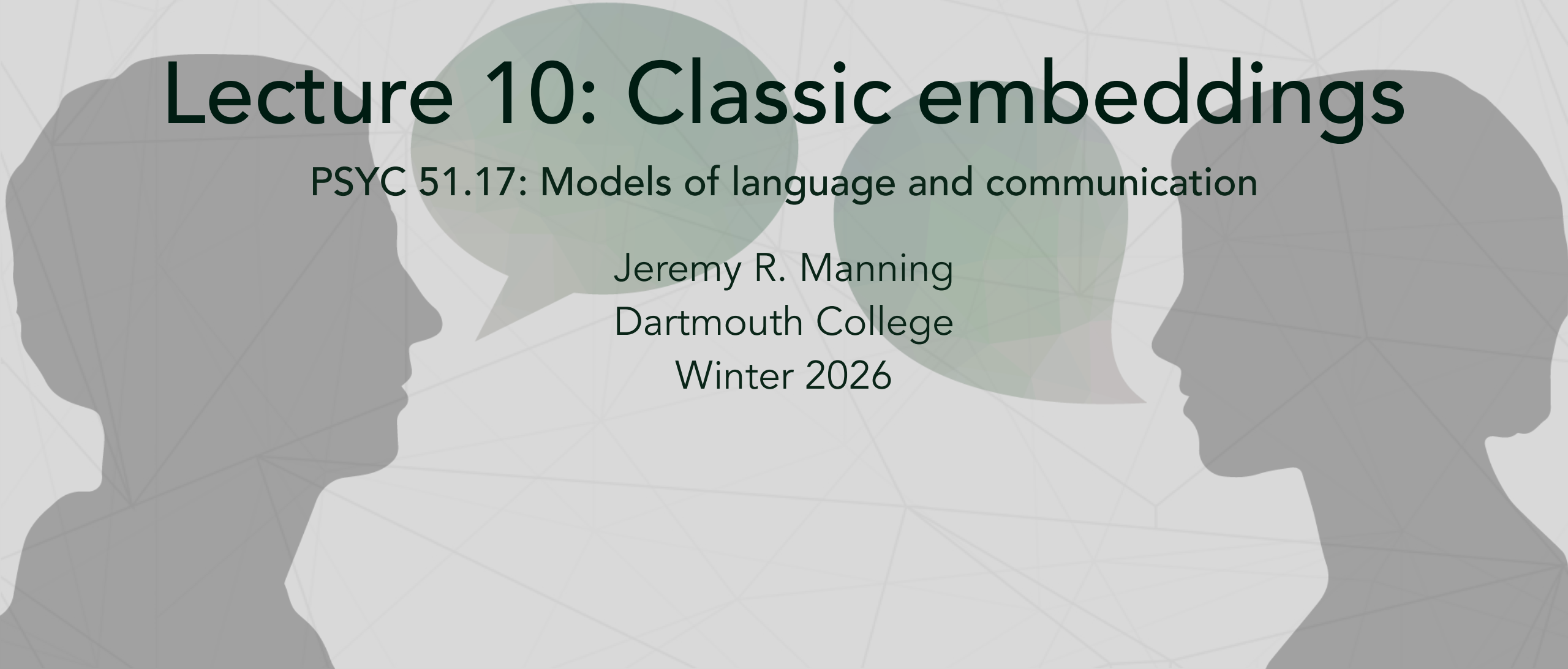


Learning objectives

By the end of this lecture, you will

1. Understand the distributional hypothesis and why it matters
2. Implement Latent Semantic Analysis (LSA) with SVD
3. Apply Latent Dirichlet Allocation (LDA) for topic modeling
4. Compare classic embedding methods and their trade-offs

Central idea

"You shall know a word by the company it keeps" — J.R. Firth (1957)

How do we represent the meaning of words computationally?

Traditional approaches: labor-intensive and expensive to scale!

- Symbolic representations (e.g., dictionaries, taxonomies)
- Manual feature engineering (e.g., counting letters, syllables, POS tags)
- Taxonomies (e.g., WordNet)

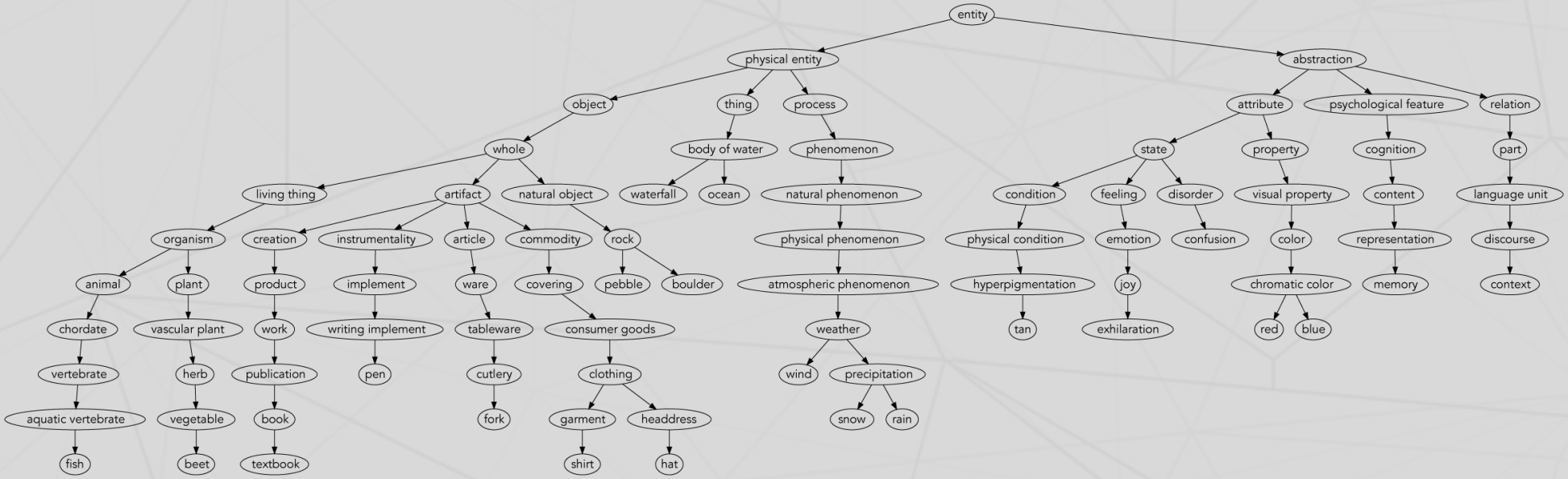
Distributional approaches: data-driven, scalable, unsupervised

- Learn from data (large text corpora)
- Context-based (words in context)
- Let the data tell us what words mean!

Example "traditional" approach: Princeton WordNet

Further reading

Miller (1995, *Communications of the ACM*) WordNet: A Lexical Database for English.



The distributional hypothesis

Definition

Words that appear in similar contexts tend to have similar meanings.

Example 1

- "The **cat** sat on the mat"
- "The **dog** sat on the mat"
- "The **kitten** sat on the mat"

→ *cat*, *dog*, *kitten* are probably semantically related

Example 2

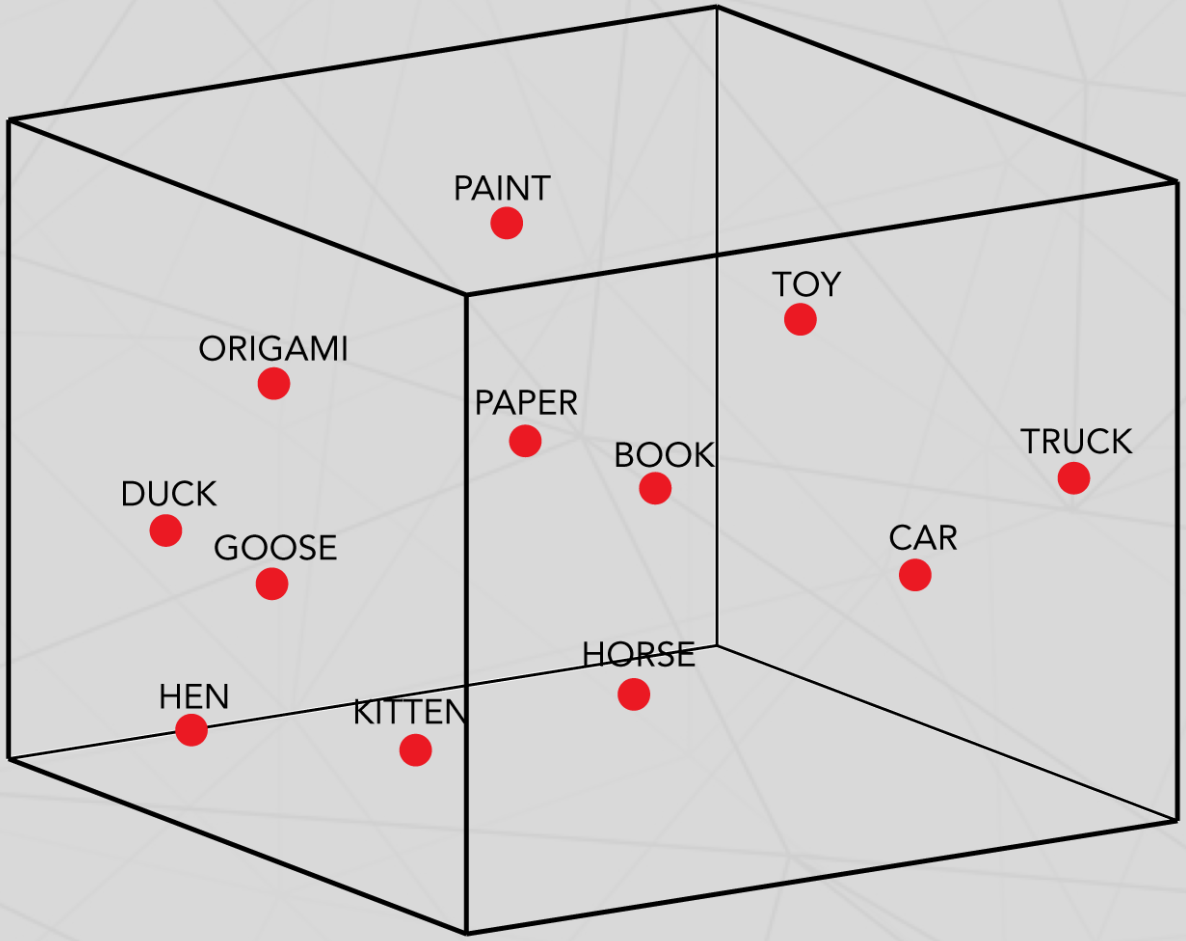
- "The **computer** processes **data** quickly by using advanced **algorithms**."

→ *computer*, *data*, *algorithms* are probably semantically related

Implication

We can learn word meaning from co-occurrence patterns alone!

From **words** to **vectors**: represent each word as a point in a high-dimensional space



How can we construct these high-dimensional word vectors?

- If words that appear in similar contexts have similar meanings, we can represent each word by the contexts it appears in:
  - We could count how often each word appears near other words (co-occurrence counts)
  - We could count how often each word appears in each document in a corpus (term-document matrix)

Latent Semantic Analysis (LSA): the OG distributional word representation

Core idea

Words that occur in similar documents have similar meanings. We can represent each word by the documents it appears in.

Term-document matrix: how many times does each word appear in each document?

Document	apple	banana	computer	data	algorithm
Doc 1	3	2	0	0	0
Doc 2	0	0	4	5	2
Doc 3	1	1	1	0	0
Doc 4	0	0	2	3	4
Doc 5	2	0	4	0	0
Doc 6	0	3	0	0	0

Latent Semantic Analysis (LSA): the OG distributional word representation

Further reading

Deerwester et al. (1990, *Journal of the American Society for Information Science*) Indexing by Latent Semantic Analysis.

- Building the term-document matrix gives us a high-dimensional representation of each word (one dimension per document)
- It also gives us a high-dimensional representation of each document (one dimension per word)
- **But:** real corpora have thousands of documents and tens of thousands of words!
  - Some dimensions (documents, words) will be noisy; we should try to remove them
  - Some dimensions might be highly correlated with each other; we should try to combine them to improve reliability and efficiency
  - We can use dimensionality reduction to find a lower-dimensional representation that captures the important structure in the data

Formal definition of LSA

The LSA algorithm

1. Build term-document matrix X
2. Perform SVD: $X = U\Sigma V^T$
3. Keep top k dimensions
4. Use U_k as word embeddings

SVD (Singular Value Decomposition)

$X = U\Sigma V^T$ decomposes matrix X into the product of three matrices:

- U: word-topic associations (number of words × number of topics)
- Σ: topic strengths (diagonal matrix)
- V^T: topic-document associations (number of topics × number of documents)

Why it matters

Learning word representations (vectors) turns the abstract problem of defining "meaning" into a concrete linear algebra problem!

LSA in Python

```python
from sklearn.decomposition import TruncatedSVD
from sklearn.feature_extraction.text import CountVectorizer

documents = [
    "The cat sat on the mat",
    "The dog ran in the park",
    "Machine learning is amazing",
]

# Build term-document matrix (bag-of-words)
vectorizer = CountVectorizer(max_features=5000, stop_words='english')
doc_term_matrix = vectorizer.fit_transform(documents)

# Apply LSA - reduce to 100 dimensions
# (Note: for small data, use smaller n_components)
lsa = TruncatedSVD(n_components=100, random_state=42)
doc_embeddings = lsa.fit_transform(doc_term_matrix)
```

Finding similar words with LSA

```python
from sklearn.metrics.pairwise import cosine_similarity

# Get word embeddings
vocab = vectorizer.get_feature_names_out()
word_embeddings = lsa.components_.T

def find_similar(word, top_n=5):
    idx = list(vocab).index(word)
    word_vec = word_embeddings[idx].reshape(1, -1)
    sims = cosine_similarity(word_vec, word_embeddings)[0]
    top_indices = sims.argsort()[-top_n-1:-1][::-1]
    return [(vocab[i], f"{sims[i]:.3f}") for i in top_indices]

print(find_similar("computer"))
# [('software', 0.82), ('program', 0.79), ('system', 0.71), ...]
```

Where does LSA fall short?

- Consider the word "bank":
  - "He deposited money in the bank."
  - "The river overflowed its bank."
  - "He had to bank the airplane to the left."
- LSA will create a single vector for "bank" that mixes these meanings
- In general: LSA assumes linear relationships, which may not capture complex semantics

Latent Dirichlet Allocation (LDA)

Further reading

Blei, Ng, & Jordan (2003, *Journal of Machine Learning Research*) Latent Dirichlet Allocation.

A generative model for documents

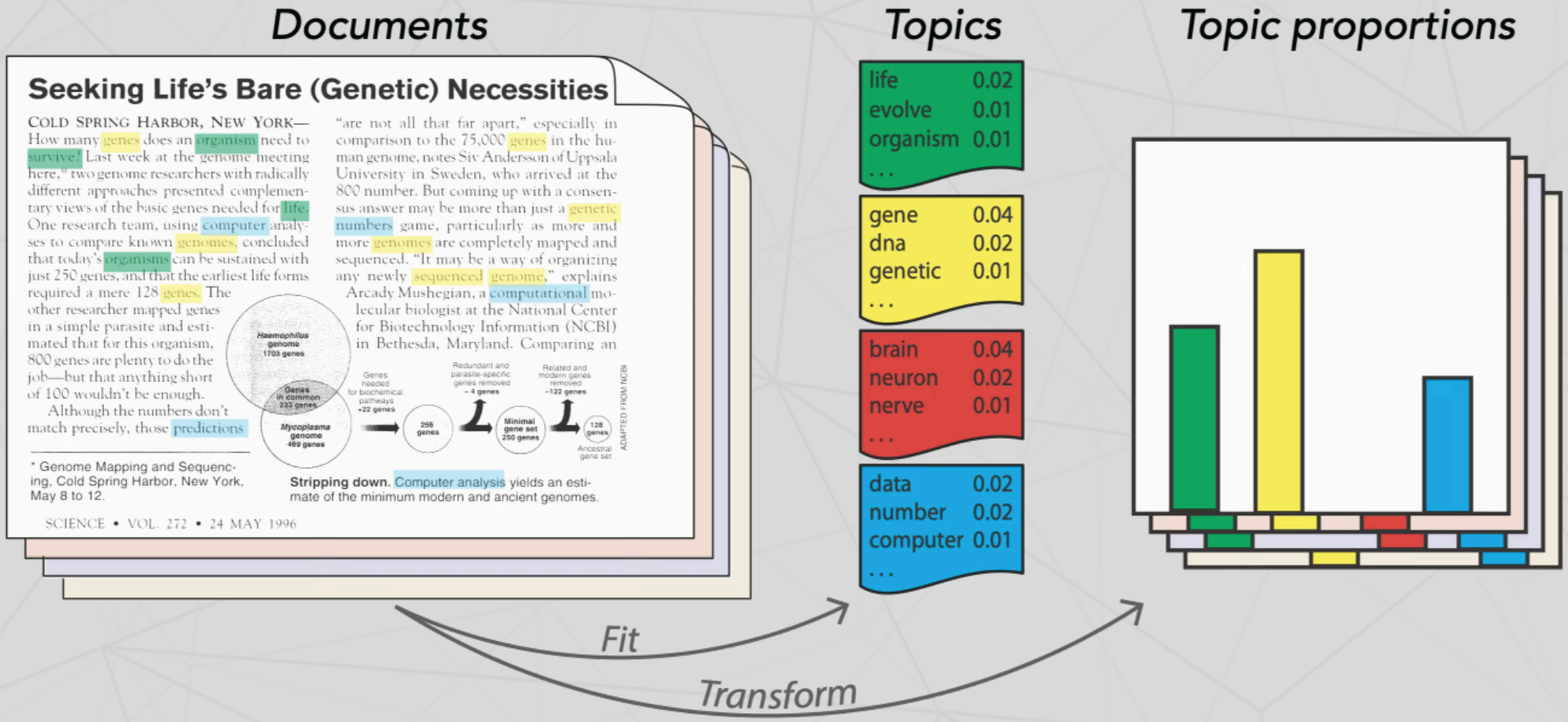
Documents are generated by a mixture of topics, where each topic is a distribution over words. For each document $\mathbf{w}$ in the corpus:

1. Choose $N \sim \text{Poisson}(\xi)$ (number of words in document)
2. Choose topic proportions $\theta \sim \text{Dirichlet}(\alpha)$ (mixture of topics)
3. For each word w_n:
   - Choose a topic $z_n \sim \text{Multinomial}(\theta)$ (topic assignment for this word)
   - Choose a word w_n from $p(w_n|z_n, \beta)$ (a multinomial probability conditioned on the topic; β is the topic-word matrix)

Why this is useful

By writing down a "recipe" for how documents are generated, we can invert the process to infer the hidden topics and their distributions from observed documents! It lets us capture the idea that documents can cover multiple topics, and that individual words can be associated with multiple topics.

LDA: cartoon


Documents
Topics
Topic proportions
Seeking Life's Bare (Genetic) Necessities
COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.
Although the numbers don't match precisely, those predictions
"are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an
Haemophilus genome 1703 genes
Genes in common 233 genes
Mycoplasma genome 482 genes
Genes needed for biochemical pathways +22 genes
256 genes
Redundant and parasite-specific genes removed −4 genes
Minimal gene set 250 genes
Related and modern genes removed −122 genes
128 genes
Ancestral gene set
ADAPTED FROM NCBI
* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.
Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.
SCIENCE • VOL. 272 • 24 MAY 1996
life 0.02
evolve 0.01
organism 0.01
...
gene 0.04
dna 0.02
genetic 0.01
...
brain 0.04
neuron 0.02
nerve 0.01
...
data 0.02
number 0.02
computer 0.01
...
Fit
Transform

LDA in Python

```python
from sklearn.decomposition import LatentDirichletAllocation
from sklearn.feature_extraction.text import CountVectorizer

# Create bag-of-words
vectorizer = CountVectorizer(max_features=1000, stop_words='english')
doc_term_matrix = vectorizer.fit_transform(documents)
vocab = vectorizer.get_feature_names_out()

# Fit LDA
lda = LatentDirichletAllocation(n_components=5, random_state=42)
doc_topics = lda.fit_transform(doc_term_matrix)

# Print top words per topic
for idx, topic in enumerate(lda.components_):
    top_words = [vocab[i] for i in topic.argsort()[-7:]]
    print(f"Topic {idx}: {', '.join(top_words)}")
```

Discussion: are distributional word representations enough to capture meaning?

Yes!

- Captures semantic similarity (e.g., synonyms)
- Works well in practice (e.g., information retrieval)
- Aligns with linguistic theory (e.g., distributional hypothesis)
- Scalable to large corpora

But also...no!

- Text-only (no grounding, missing embodied experience)
- No common sense reasoning (e.g., "the sky is blue")
- Reflects biases in data
- The approaches we've seen today are "bag-of-words" methods (ignore word order)

Key ideas from today

1. **Distributional Hypothesis:** words in similar contexts have similar meanings
2. **LSA:** SVD on term-document matrix— dense embeddings that capture global structure
3. **LDA:** probabilistic topic modeling— interpretable topics and word distributions

Questions?

Email me

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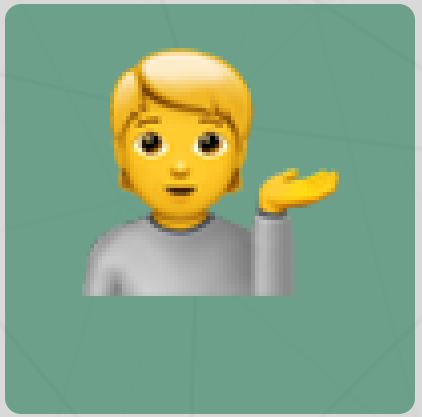

Come to office hours

Up next...

We'll kick off **week 4** with a lecture on **neural embeddings** (Word2Vec, GloVe, FastText)! Also remember that **Assignment 2** is due on **Monday at 11:59 PM**!